Griffiths Chapter 10: Gauge Theories

A First-Exposure Summary of Sections 10.8–10.9

PHYS 7502 Reading Guide

The Problem These Sections Solve

Local gauge invariance *requires* gauge fields to be massless (the Proca mass term breaks gauge invariance). But the $W^{\pm}$ and Z^0 bosons are massive (~ 80–91 GeV/c^2).

How can we give mass to gauge bosons without destroying gauge invariance?

Answer: Spontaneous symmetry breaking + the Higgs mechanism. Section 10.8 sets up the idea; Section 10.9 delivers the punchline.

1 Section 10.8 — Spontaneous Symmetry Breaking

What is Spontaneous Symmetry Breaking?

Term Refresher: Spontaneous Symmetry Breaking

A symmetry is **spontaneously broken** when the *Lagrangian* possesses the symmetry, but the *ground state* (vacuum) does not.

The symmetry isn't destroyed — it's **hidden** by the system's choice of ground state.

Everyday Analogy: The Bent Ruler

Take a thin plastic ruler and squeeze both ends together. It will buckle — but it could buckle *left* or *right* with equal probability.

The physics is perfectly left–right symmetric, but once the ruler picks a direction, the symmetry is broken. The Lagrangian is symmetric; the ground state is not.

Now replace the ruler with a *knitting needle* (a rod): it can buckle in *any* direction (360° of options). This is the continuous symmetry case, and it produces far richer physics.

A Toy Model: Two Scalar Fields with Continuous Symmetry

Griffiths constructs a Lagrangian with two real scalar fields ϕ_1 and ϕ_2:

$$L = \tfrac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_1) + \tfrac{1}{2}(\partial_\mu \phi_2)(\partial^\mu \phi_2) + \tfrac{1}{2}\mu^2(\phi_1^2 + \phi_2^2) - \tfrac{1}{4}\lambda^2(\phi_1^2 + \phi_2^2)^2 \tag{10.115}$$

Watch the sign of μ^2!

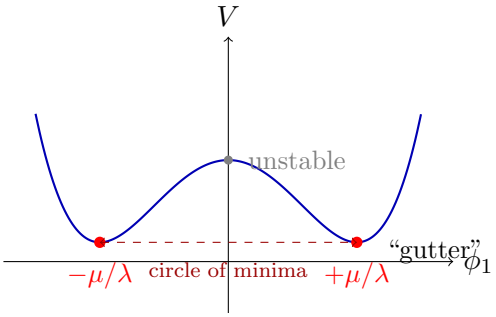
Notice the $+\mu^2$ in front of the $(\phi_1^2+\phi_2^2)$ term. This is the "wrong" sign compared to a normal mass term (which would be $-\mu^2$). The wrong sign is what makes the potential interesting.

This Lagrangian depends only on $\phi_1^2 + \phi_2^2$, so it is invariant under rotations in (ϕ_1, ϕ_2) space — an $SO(2)$ symmetry (equivalently, $U(1)$).

The "Mexican Hat" Potential

The potential energy is:

$$V(\phi_1, \phi_2) = -\tfrac{1}{2}\mu^2(\phi_1^2 + \phi_2^2) + \tfrac{1}{4}\lambda^2(\phi_1^2 + \phi_2^2)^2 \tag{10.116}$$



The minima lie on a **circle** of radius μ/λ in (ϕ_1, ϕ_2) space:

$$\phi_1^2 + \phi_2^2 = \frac{\mu^2}{\lambda^2} \equiv v^2 \tag{10.117}$$

The point $\phi_1 = \phi_2 = 0$ is a *local maximum* (unstable). The physical vacuum must sit somewhere on the circle of minima.

Choosing a Vacuum and Expanding Around It

Every point on the circle is equally valid. Pick one — say:

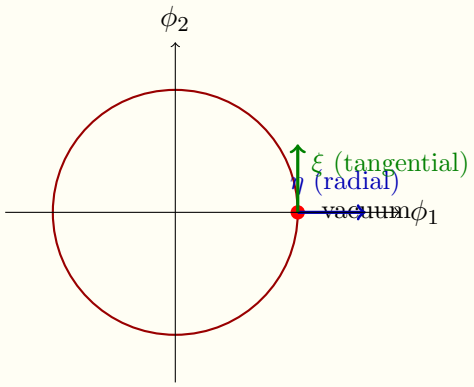
$$\phi_1 = \mu/\lambda, \qquad \phi_2 = 0 \tag{10.118}$$

Now define new fields η and ξ as fluctuations about this vacuum:

$$\eta \equiv \phi_1 - \mu/\lambda, \qquad \xi \equiv \phi_2 \tag{10.119}$$

What do η and ξ represent physically?

Stare at the Mexican hat potential from above (looking down at the circle of minima):



- η = radial fluctuation (climbing the wall of the potential). This costs energy → **massive** particle.
- ξ = tangential fluctuation (rolling around the gutter). This costs *no* energy → **massless** particle (Goldstone boson).

The Rewritten Lagrangian

Substituting $\phi_1 = \eta + \mu/\lambda$ and $\phi_2 = \xi$ into (10.115) gives:

$$L = \underbrace{\tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \mu^2\eta^2}_{\text{massive } \eta,\ \text{mass } m_\eta = \sqrt{2}\,\mu\hbar/c} + \underbrace{\tfrac{1}{2}(\partial_\mu\xi)(\partial^\mu\xi)}_{\text{massless } \xi} + \text{interaction terms} \tag{10.120}$$

Goldstone's Theorem

Spontaneous breaking of a **continuous global** symmetry always produces one or more **massless scalar particles** — the **Goldstone bosons**.

In this example, the $SO(2)$ symmetry is broken by the choice of vacuum, and the field ξ (tangential excitations along the gutter) is the Goldstone boson.

Why is this a disaster?

We wanted to use spontaneous symmetry breaking to give mass to gauge bosons. But instead it introduces a *new* massless particle (the Goldstone boson) that doesn't exist in nature.

No spin-0 massless particle has ever been observed. This seems to be a dead end... but Section 10.9 rescues everything.

2 Section 10.9 — The Higgs Mechanism

The Twist

What happens when you combine spontaneous symmetry breaking with **local** gauge invariance (instead of just global)?

Answer: The Goldstone boson **disappears**, and the gauge field **acquires mass**. This is the Higgs mechanism.

Step 1: Rewrite the Two Real Fields as One Complex Field

Combine ϕ_1 and ϕ_2 into a single complex scalar field:

$$\phi = \phi_1 + i\phi_2, \qquad \phi^*\phi = \phi_1^2 + \phi_2^2 \tag{10.123–124}$$

The Lagrangian (10.115) becomes:

$$L = \tfrac{1}{2}(\partial_\mu\phi^*)(\partial^\mu\phi) + \tfrac{1}{2}\mu^2(\phi^*\phi) - \tfrac{1}{4}\lambda^2(\phi^*\phi)^2 \tag{10.125}$$

The $SO(2)$ rotational symmetry in (ϕ_1, ϕ_2) space is now a $U(1)$ phase symmetry: $\phi \to e^{i\theta}\phi$. This is exactly the symmetry we know how to gauge!

Step 2: Make It Locally Gauge Invariant

Promote θ to $\theta(x)$ and introduce a gauge field A_μ via the covariant derivative:

$$D_\mu = \partial_\mu + \frac{iq}{\hbar c} A_\mu \tag{10.128}$$

The locally gauge-invariant Lagrangian is:

$$L = \tfrac{1}{2}(D_\mu\phi)^*(D^\mu\phi) + \tfrac{1}{2}\mu^2(\phi^*\phi) - \tfrac{1}{4}\lambda^2(\phi^*\phi)^2 - \frac{1}{16\pi}F^{\mu\nu}F_{\mu\nu} \tag{10.129}$$

At this stage A_μ is massless (as required by gauge invariance).

Step 3: Break the Symmetry

Just as before, the vacuum sits at $|\phi| = \mu/\lambda \equiv v$, and we expand around it. Define:

$$\eta \equiv \phi_1 - v, \qquad \xi \equiv \phi_2 \tag{10.130}$$

Rewrite the Lagrangian in terms of η, ξ, and A_μ. After the dust settles (Problem 10.25), the result is:

$$\begin{aligned} L = {} & \underbrace{\tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \mu^2\eta^2}_{\text{massive scalar } \eta} + \underbrace{\tfrac{1}{2}(\partial_\mu\xi)(\partial^\mu\xi)}_{\text{massless Goldstone boson } \xi} \\ & + \underbrace{-\frac{1}{16\pi}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}\left(\frac{qv}{\hbar c}\right)^2 A^\mu A_\mu}_{\text{massive gauge field!}} \\ & + \{\text{various interaction terms}\} + \underbrace{\frac{qv}{\hbar c}(\partial_\mu\xi)\,A^\mu}_{\text{suspicious } \xi\text{-}A \text{ mixing}} \end{aligned} \tag{10.131}$$

Three remarkable things just happened

1. The gauge field A_μ has **acquired a mass**:

$$m_A = \frac{2qv}{\hbar c} = \frac{2q\mu}{\hbar c\lambda} \tag{10.132}$$

   This came from the $\phi^*\phi\, A_\mu A^\mu$ term in the Lagrangian. Before symmetry breaking, this was just a coupling; after the vacuum shifts by v, it generates the Proca mass term $\frac{1}{2}m_A^2\, A_\mu A^\mu$.
2. The Goldstone boson ξ is still here — we haven't gotten rid of it yet.
3. There is a suspicious term $(\partial_\mu\xi)A^\mu$ that mixes ξ and A_μ. A term bilinear in two different fields means we haven't correctly identified the physical particles.

Step 4: Gauge Away the Goldstone Boson

The Key Insight

The Lagrangian (10.129) is invariant under local gauge transformations. We haven't yet *used* this freedom. A clever choice of gauge can eliminate ξ entirely!

Under a $U(1)$ phase rotation $\phi \to e^{i\theta}\phi$ with the specific choice:

$$\theta = -\tan^{-1}(\phi_2/\phi_1) \tag{10.135}$$

the field ϕ is rotated to be entirely *real* ($\phi_2' = 0$, hence $\xi = 0$). This is called the **unitary gauge**.

In this gauge, the Lagrangian reduces to:

The Higgs Mechanism Lagrangian (Unitary Gauge)

$$L = \underbrace{\tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \mu^2\eta^2}_{\text{massive Higgs boson } \eta} + \underbrace{-\frac{1}{16\pi}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}\left(\frac{qv}{\hbar c}\right)^2 A^\mu A_\mu}_{\text{massive gauge boson } A_\mu} + \{\text{interactions}\} \tag{10.136}$$

The Goldstone boson ξ is gone. The gauge field A_μ is massive. And we have a new massive scalar particle η — the **Higgs boson**.

The Degree-of-Freedom Count

Before symmetry breaking:
- 2 real scalar fields (ϕ_1, ϕ_2) = 2 degrees of freedom
- 1 massless gauge field A_μ = 2 degrees of freedom (transverse polarizations)
- Total: **4**

After symmetry breaking (unitary gauge):
- 1 massive Higgs boson (η) = 1 degree of freedom
- 1 massive gauge field A_μ = 3 degrees of freedom (2 transverse + 1 longitudinal)
- Total: **4** ✓

No degrees of freedom were created or destroyed. The Goldstone boson was "eaten" by the gauge field, which gained a longitudinal polarization and became massive.

Where Did the Gauge Boson Mass Come From?

Tracing the mass term

The locally gauge-invariant Lagrangian (10.129) contains the term:

$$(D_\mu\phi)^*(D^\mu\phi) \supset \left(\frac{q}{\hbar c}\right)^2 \phi^*\phi\, A_\mu A^\mu$$

Before symmetry breaking, $\langle\phi\rangle = 0$ and this is just an interaction vertex (four lines meeting: ϕ^*, ϕ, A_μ, A^μ).

After symmetry breaking, $\langle\phi_1\rangle = v \neq 0$. The field $\phi_1 = v + \eta$, so $\phi^*\phi$ acquires a constant piece v^2. This turns the interaction vertex into a **mass term**:

$$\left(\frac{q}{\hbar c}\right)^2 v^2\, A_\mu A^\mu \quad \longrightarrow \quad \frac{1}{2} m_A^2\, A_\mu A^\mu$$

The gauge field mass is *not* put in by hand — it *emerges* from the vacuum expectation value of the scalar field.

Summary: What Each Piece Became

Before SSB	**After SSB**	**Physical Particle**
ϕ_1 (radial mode)	massive scalar η	**Higgs boson** ($m_\eta = \sqrt{2}\,\mu\hbar/c$)
ϕ_2 (tangential mode)	"eaten" by A_μ	(longitudinal polarization of A_μ)
A_μ (massless, 2 polarizations)	massive, 3 polarizations	**massive gauge boson**

Application to the Standard Model

How this works for the $W^{\pm}$ and Z^0

In the Standard Model, the Higgs mechanism is applied to the $SU(2)_L \times U(1)_Y$ gauge symmetry of the electroweak interaction. The Higgs field is a *doublet* of complex scalars (four real fields total), with a Mexican-hat potential.

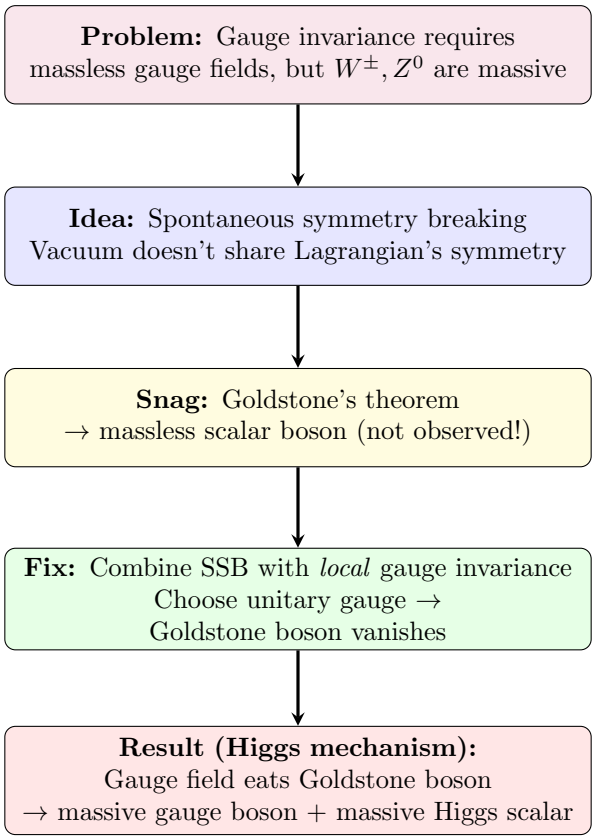
After spontaneous symmetry breaking:

- 3 Goldstone bosons are eaten by the W^+, W^-, and Z^0, giving them mass ($m_W \approx 80\ \text{GeV}/c^2$, $m_Z \approx 91\ \text{GeV}/c^2$).
- 1 remaining scalar degree of freedom becomes the **Higgs boson** h ($m_h \approx 125\ \text{GeV}/c^2$, discovered at CERN in 2012).
- The photon remains **massless** (the $U(1)_{\text{em}}$ symmetry is unbroken).

The gauge boson masses are set by the **vacuum expectation value** $v \approx 246$ GeV and the gauge coupling constants:

$$m_W = \tfrac{1}{2} g_W v, \qquad m_Z = \frac{v}{2}\sqrt{g_W^2 + g'^2}, \qquad m_\gamma = 0$$

Master Summary: The Logic of Sections 10.8–10.9



The Big Takeaway

The Higgs mechanism is the marriage of two ideas:

1. **Local gauge invariance** (Sections 10.3–10.5): forces the existence of gauge fields, but requires them to be massless.
2. **Spontaneous symmetry breaking** (Section 10.8): the vacuum picks a ground state that hides the Lagrangian's symmetry, producing Goldstone bosons.

When you combine them, a miracle occurs: the Goldstone boson is absorbed into the gauge field as its longitudinal polarization, and the gauge field becomes massive — all without ever explicitly breaking gauge invariance in the Lagrangian.

The Lagrangian is still gauge-invariant. The symmetry is hidden, not destroyed.

This is the theoretical foundation for the masses of the $W^{\pm}$ and Z^0 in the Standard Model, confirmed by the discovery of the Higgs boson at the LHC in 2012.